

RES-5 sunset-norm certificate: the leading skeleton IS SC-SCOPE (thin);

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The a0-skeleton note reduced RES-5 to $|\Delta\Gamma_2^{\text{pd}}| \leq c a_0 \Delta F_{\text{margin}}$ and estimated $c a_0 \sim 0.002$. The operator flagged the normalization map (the SC-SCOPE quantity is a free-energy difference, not a self-energy norm). This certificate corrects that conflation, maps the residual onto the SC-SCOPE certified joint, and gives the honest verdict: RES-5 survives at the SAME thin grade as SC-SCOPE. Scope: the normalization map; the precise identity and the tail bound are the residual.

2. Correction: the certificate quantity is a free-energy ratio

The a0-skeleton estimate $c a_0 \sim 0.49 \cdot a_0 \cdot (\|\Sigma_2^{\text{pd}}\|/\Delta F_{\text{margin}})$ used $\|\Sigma_2^{\text{pd}}\|/\Delta F_{\text{margin}} \sim 0.04$ – but that 0.04 is the FREE-ENERGY (sunset) ratio, not the SELF-ENERGY norm. The two are related by $\Delta\Gamma_2^{\text{pd}} = \|\Sigma_2^{\text{pd}}\| \|\delta G_*^{\text{pd}}\|$, so quoting the free-energy ratio AND multiplying by $\|\delta G_*^{\text{pd}}\| = O(a_0)$ double-counts the a_0 . The CERTIFICATE QUANTITY is directly the free-energy ratio

$$\frac{|\Delta\Gamma_2^{\text{pd}}|}{\Delta F_{\text{margin}}},$$

whose leading (sunset / 2-loop) value is what SC-SCOPE certified – the THIN joint, not 0.2%.

3. The normalization map: the SC-SCOPE joint IS the leading certificate

SC-SCOPE certified the endpoint joint

$$\text{joint} = \frac{\text{MARGIN}}{C_2 + C_{\text{sunset}} + C_{\text{quartic}}} = \times 1.040 \text{ (certified)} \longrightarrow \times 1.13 \text{ (sunset cap)},$$

both > 1 . This is PRECISELY the statement that the third-cumulant (the LEADING 2PI skeleton, sunset + quartic) does NOT overturn the one-loop selection: the leading $\Delta\Gamma_2^{\text{pd}}$ certificate. It is THIN (the joint exceeds unity by 4–13%), sunset-limited.

4. Higher skeletons: screened sub-dominant

The $\ell \geq 3$ skeletons are bounded by the 2PI screened response (res5-a0-skeleton-sensitivity): each is the previous order times the screened factor $1/(1+g) = 0.49$, so the geometric tail

$$\sum_{\ell \geq 3} |\Delta\Gamma_{\ell}^{\text{pd}}| \leq |\Delta\Gamma_{2\text{-loop}}^{\text{pd}}| \cdot \frac{1}{1-0.49} \approx 2 |\Delta\Gamma_{2\text{-loop}}^{\text{pd}}|.$$

The total skeleton correction is therefore at most $\sim 2\times$ the (thin) sunset, and the selection survives the matched-order-to-exact with a margin of the same thin order as SC-SCOPE.

5. Verdict and residual

T4 STRONG EVIDENCE (thin). RES-5 survives the matched-order-to-exact: the leading skeleton is the SC-SCOPE joint ($\times 1.04\text{--}1.13 > 1$, certified), and the higher skeletons are screened sub-dominant ($\leq \times 2$ the leading). The matched-order-to-exact selection holds at the SAME thin grade as SC-SCOPE. **OPEN (the residual):** (i) the rigorous identity mapping the SC-SCOPE joint bookkeeping (layer-margin / cost) to $\Delta\Gamma_2^{\text{pd}}/\Delta F_{\text{margin}}$ (the skeleton free-energy ratio); (ii) the rigorous higher-skeleton tail bound (the screened-geometric estimate made an inequality). No tier flip: B1-RH-ENUM T6 CONDITIONAL on {H-LAYER}; the thin grade matches SC-SCOPE's standing as a named hypothesis.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) “The $c a_0 \sim 0.002$ was simply wrong.” UPHELD and corrected: it double-counted a_0 (free-energy ratio $\times$ response). The honest value is the SC-SCOPE thin joint ($\times 1.04$), not 0.2%. The session's discipline (operator point 2) caught it.

(β) “The SC-SCOPE-joint $\rightarrow \Delta\Gamma_2^{\text{pd}}$ map is asserted, not proved.” VALID: the two bookkeepings (layer-margin/cost vs one-loop-selection/skeleton) coincide structurally (both are the third-cumulant correction to the selection), but the exact normalization identity is the residual.

(γ) “Then RES-5 is no better than SC-SCOPE.” PARTLY-UPHELD: the LEADING skeleton is exactly SC-SCOPE (thin); the NEW content is the 2PI structure controlling the HIGHER skeletons (common-mode mechanism + screened response), which SC-SCOPE did not address. RES-5 = SC-SCOPE (leading) + screened higher tail (this session).

Quantitative sanity check (mandatory): *correction* – 0.002 was free-energy $\times$ response (double-counted a_0); *leading* – SC-SCOPE joint $\times 1.04\text{--}1.13 > 1$; *tail* – screened geometric $\leq \times 2$ leading; *grade* – thin, sunset-limited, = SC-SCOPE.

7. Result footer

Result ID:	B1-RH-ENUM / RES-5 sunset-norm certificate
Precise statement:	CORRECTION: the a0-skeleton $c a_0 \sim 0.002$ double-counted a_0 (free-energy ratio $\times$ response). The certificate quantity is $ \Delta\Gamma_2^{\text{pd}} /\Delta F_{\text{margin}}$; its leading (sunset) value IS the SC-SCOPE certified joint = $\text{MARGIN}/(\text{C2}+\text{C}_{\text{sunset}}+\text{C}_{\text{quartic}}) = \times 1.040 \rightarrow \times 1.13 (>1: \text{third cumulant does not})$

overturn the selection). Higher skeletons screened sub-dominant ($\text{tail} \leq \text{leading}/(1-0.49) \sim x^2$). RES-5 survives the matched-order-to-exact at STRONG EVIDENCE, THIN -- same grade as SC-SCOPE. New content vs SC-SCOPE: the 2PI common-mode + screened-response control of the HIGHER skeletons.

Scope: The normalization map; anchor μ^2 . The rigorous identity and the tail bound are OPEN.

Dependencies: res5-a0-skeleton-sensitivity (screened response, corrected here); res5-higherloop-commonmode-monotonicity (2PI common mode); SC-SCOPE (the joint x1.04-x1.13); Math424-AddA.

Evidence grade: ANALYTIC + EXECUTED (res5_sunset_norm_map.py 4/4) + INHERITED (SC-SCOPE joint).

Reproduction command: python codes/vacuum/res5_sunset_norm_map.py

Expected output: SC-SCOPE joint x1.04-x1.13>1; screened tail x1.97; 4/4.

Falsification gate: A normalization identity giving $\Delta \Gamma_2^{\text{pd}}/\Delta F_{\text{margin}} \geq 1$ (selection overturned); or a higher-skeleton tail NOT screened (g-amplified).

Tier before / after: No tier flip. B1 T6 on {H-LAYER}. RES-5: matched-order-to-exact STRONG EVIDENCE thin (= SC-SCOPE leading + 2PI screened higher tail).

No-overclaim statement: This does NOT close RES-5 and does NOT flip B1. It corrects a self-energy/free-energy conflation, maps the residual onto the SC-SCOPE thin joint, and adds the 2PI screened control of the higher skeletons. The grade is THIN (= SC-SCOPE); the rigorous normalization identity and tail bound remain. B1 stays T6 on {H-LAYER}.

Next required action: Prove the SC-SCOPE-joint $\rightarrow \Delta \Gamma_2^{\text{pd}}/\Delta F_{\text{margin}}$ identity and the rigorous screened higher-skeleton tail bound; then RES-5 $\rightarrow$ controlled-error theorem at the thin (SC-SCOPE) grade.